

Macroeconomic objectives

Economic analysis is concerned with the means of achieving particular economic objectives. The choices of the objectives – how people want economic resources to be used in order to satisfy their wants – depend upon the economic situation of the economy. While governments differ in the emphasis they give to particular objectives and in the ways in which they try to achieve them, there seems to be broad general agreement on the main aims. These are a high and stable level of employment, price stability, a satisfactory balance of payments, an acceptable rate of economic growth and an equitable distribution of income and wealth. Of these possible economic goals, only few are likely to be central and of major importance to the authorities; and what is regarded central can change over time.

Maintaining high employment level is one of the most important aims of government economic policy. When employment level is high, output and hence living standard are also usually high. Most governments prefer to keep unemployment at a low level, partly for economic reasons but probably the main motives are political and social. The government can lose support of the people if unemployment is high and large numbers of the unemployed can cause social unrest. The economy suffers a loss of potential output due to high unemployment and hence actual living standards are below potential living standards. Governments in recent years have been very concerned about employment level and pursue expansionary fiscal policy as a tool to achieve higher levels of employment.

Another important aim of governments is to manage the economy in such a way that economic growth will be fostered. Short-term economic growth is termed as actual growth which is the increase in real GDP or GNP of the country. Long-term economic growth is termed as potential growth which is the increase in the productive potential of the economy.

As a long-term goal, most governments seem to like economic growth try to adopt policies to promote it. A higher rate of economic growth means that we can enjoy a higher standard of living gives us a greater ability to improve resource allocation and also provides the surplus out of which we can take better care of environment. Against this the process of economic growth is frequently bad for the environment, particularly by causing pollution of the air and water, as well as noise.

In order to accelerate economic growth, government influence the level of private investments through monetary and fiscal tools like grants, tax concessions, interest free loans etc, other investments in human capital, encouraging the technical progress and so on.

Achieving an equal distribution of income and wealth is yet another important aim of government economic policy. The distribution of income in an economy is the proportion of income earned by a given proportion of population, the income is said to be equally distributed. On the other hand, if a larger proportion of income is earned by a relatively smaller proportion of population, it is said to be unevenly distributed. Governments seek to narrow the gap between the rich and the poor by

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reducing the incomes of those at the upper end and raising the incomes of those at the lower end. This is done by adopting progressive tax system.

A satisfactory BOP is also an important aim of a government, so it wishes either to balance the current or perhaps aim for a small surplus. Persistent deficits in BOP can have very serious consequences on an economy. A deficit in BOP will be financed either through foreign currency reserves through borrowing which will either reduce foreign currency reserves or increase external debt which hampers growth and the overall standard of living falls.

Of these possible economic goals only few are likely to be central and major importance to the authorities; and what is regarded as central can change over time. It is not possible to achieve all of these economic aims at the same time, either because some goal clash with others, or the available resources are too small to allow the pursuit of more than a few of them. For such reasons, a choice must be made and the goals placed into some order, so that policies can be formed to tackle what is felt to be the core objective.

Conflicts between objectives

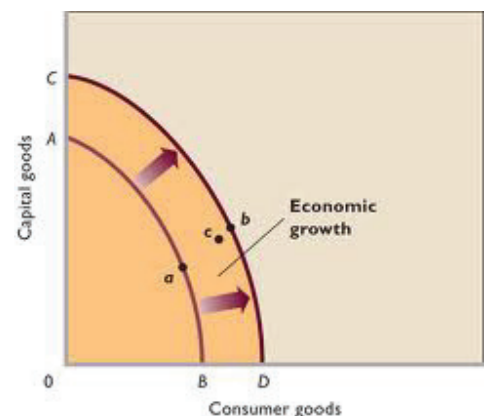
1. Healthy growth and low inflation
2. Healthy growth and a balance of payment equilibrium
3. Low unemployment (or full employment) and low inflation
4. Healthy growth and the environment
5. Healthy growth and equality

Economic growth

The distinction between actual and potential growth: Actual growth is the percentage annual increase in national output: the rate of growth in actual output. When the statistics on growth rate are published, it is actual growth they are referring to. Potential growth is the speed at which the economy could grow. It is the percentage annual increase in the economy's capacity to produce: the rate of growth in potential output. Two of the major factors contributing to potential economic growth are:

1. An increase in resources - natural resources, labor or capital
2. An increase in the efficiency with which these resources are used, through advances in technology, improved labor skills or improved organizations.

If the potential growth rate exceeds the actual growth rate, there will be an increase in spare capacity and probably an increase in unemployment: there will be a gap between potential and actual output. To close this gap, the actual growth rate has



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to increase. In the long-run however the actual growth rate will be limited to the potential growth rate.

The distinction between actual and potential growth in output can be illustrated on a production possibility curve. The PPC itself shows potential output and growth curve itself shows potential output and growth in potential output will lead to a rightward shift in the PPC.

In the short-run, actual growth rise from a fuller use of resources. This would involve using machinery more fully and reducing unemployment. For actual growth to be sustained over a number of years, however, there would also have to be an increase in potential output.

Actual output will tend to fluctuate i.e. in some years; countries will experience high rates of economic growth: the country experiences a boom. In other years, economic growth will be low or even negative: the country experiences a recession. This cycle of booms and recessions is known as the business cycle.

Causes of fluctuations in actual growth: The major determinants of variations in the rate of actual growth in the short-run are variations in the growth of AD which is the total spending on goods and services made within the country. A rapid rise in AD will create shortages which will tend to stimulate firms to increase output, thus reducing slack in the economy. Likewise, a reduction in AD will leave firms with increased stocks of unsold goods. They will therefore tend to reduce output.

Aggregate demand and actual output, therefore, fluctuate together in the short-run. A boom is associated with a rapid rise in AD; the faster the rise in AD, the higher the short-run rate of actual growth. A recession, by contrast is associated with a reduction in AD.

A rapid rise in AD, however, is not enough to ensure a continuing high level of growth over a number of years. Without an expansion potential output too, rises in actual output must eventually come to an end. Once, spare capacity has been used up, once there is full employment of labor and other resources, the rate of growth of actual output will be restricted to the rate of growth of potential output.

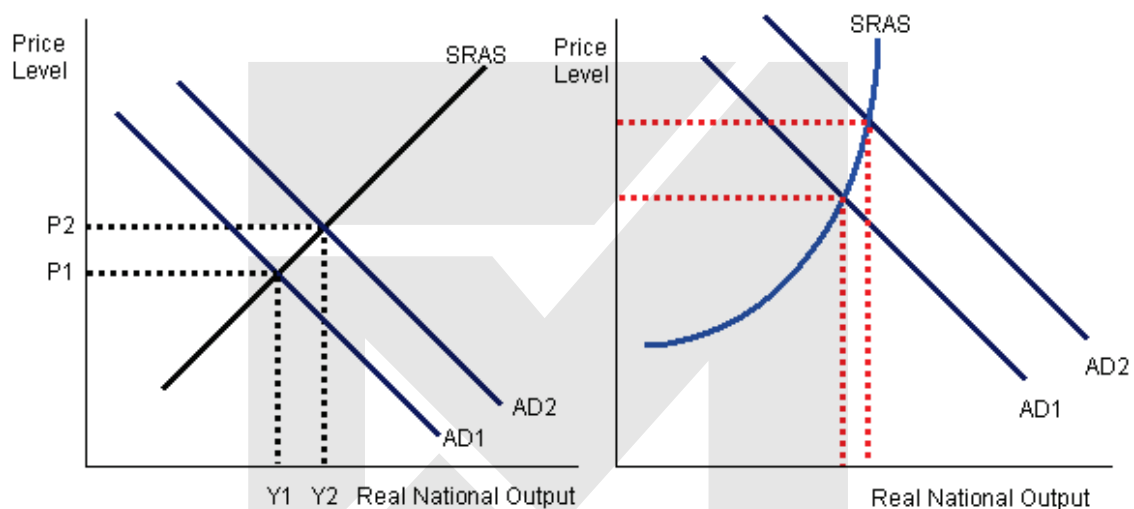
In the long run therefore there are two determinants of actual growth:

1. The growth in AD. This determines whether potential output will be realized
2. The growth in potential output

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In the figure, an increase in the AD cause the AD curve to shift to the right from AD1 to AD2 causing the excess of demand over and above the available supply of goods and services (AS) at the given price level P1. This causes the price level to rise causing it more attractive for the producers to produce more output which increases from Y1 to Y2. This increase in actual output will depend on the level of employment in the economy. If the economy is near full employment then the increase in NI will be less i.e. actual growth will slow down. Once the actual output reaches the potential output, the growth in actual output will depend on the growth in the potential output.



Causes of potential growth: There are two main determinants of potential output (a) the total amount of resources, and (b) the productivity of resources.

The nation's output depends on its stock of capital (K). An increase in this stock will increase the output. A nation's stock of capital increase as a result of positive net investment i.e. investment over and above what is necessary to replace worn output equipment. Although investment is recognized as a factor in economic growth, there is much dispute as to the way in which the two are linked. While it is true that no country has achieved a very fast growth rate without a high rate of investment, there are several examples of countries which have invested heavily and failed to achieve high growth rates. Much depends on the type of investment being undertaken. The construction of schools, houses, hospital and other forms of social capital does not have a direct influence on productive efficiency in the same way as manufacturing, commercial or agricultural investment.

Labor force is also an important determinant of the productive potential of an economy. If there will be an increase in the working population, there will be an increase in potential output. This increase in the working population may result from a higher participation rate: a larger proportion of the working population in the work or seeking work. For e.g. if a greater proportion of women decide to join the labor market, the labor force will rise.

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The total amount of natural resource is yet another important factor for the productive potential of the economy. Fertile land, a favorable climate and a good supply of minerals and fuels are obviously beneficial. However, a number of advanced countries, for e.g. Israel, do not possess a good supply of natural resources. The scope of generating growth here is usually very limited. Land is virtually fixed in quantity. Land reclamation schemes and the opening up of marginal land can add only tiny amounts to national output. Even if new raw material are discovered (e.g. oil), this will result only in short-term growth i.e. while the raw material of extraction is building up. Once the rate of extraction is at a maximum, economic growth will cease. Outputs will simply remain all the new higher level, until eventually the raw materials begin to run out. Output will then fall bank again. Apart from the quantities of all the above mentioned resources there productivities are also and important determinant for faster rates of economic growth. An increase in the productivities of the factors permanently increases the productive potential of the economy. Improvements in the methods of production brought about by introducing new technology will increase the productivities of land, labor and capital. New technology, however, is a rather vague term and it is taken to include such things as new inventions new techniques of production, improvements in the design and performance of machinery, better organization and management, more efficient factory layouts, better training facilities and more efficient systems of transport and communications. It's unrealistic to separate technical progress completely, because much of the new technology tends to be embodied in new kinds of equipment.

Costs and benefits of economic growth: There can be major advantages of economic growth, and certainly, other thing being equal, the majority of the population wants higher levels of production and consumption. But it is important to recognize the costs of economic growth. Indeed, some people regard these costs as so serious that they advocate a policy of zero economic growth.

The benefits of growth

- Increased levels of consumption: Provided economic growth outstrips population growth, it will lead to higher levels of consumption of goods and services. If human welfare is related to the level of consumption, then growth provides an obvious gain to society.
- It can help avoid other macroeconomic problems: People have aspirations of rising living standards. Without a growth in productive potential, people's demands for rising incomes are likely to lead to higher inflation, balance of payments crises, industrial disputes, etc. Growth in productive potential helps to meet these aspirations and avoid macro economic crises.
- It can make it easier to redistribute income to the poor: If incomes rise, the government can redistribute income from rich to the poor without the rich loosing. For example, as people's incomes rise, they automatically pay more taxes. These extra revenues for the government can be spending on programs to alleviate poverty. Without a continuing rise in national income, the scope for helping the poor is much more limited.

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- Society may feel that it can afford to care more for the environment: As people grow richer, they may become less preoccupied with their own private consumption and more concerned to live in a clean environment. The regulation of the pollution tends to be tougher in developed countries than in the developing countries.

The costs of growth: In practice, more consumption may not make people happier; economies may have unequal distribution of income and wealth, the environment may not be better protected. More than this, so people argue that growth may worsen these problems and create other problems besides.

- The current opportunity cost of growth: To achieve growth, firms will probably need to invest more; this will require finance which can come from higher savings, higher retained profits, or higher taxes. With either way there must be a cut in consumption. In very short run, therefore, higher growth leads to less consumption.
- Growth may simply generate extra demands: The more people have, the more they want. If this is so, more consumption may not increase people's utility at all.
- Social effects: Many people claim that an excessive pursuit of material growth by a country can also lead to a greedier, more selfish and less caring society. As society becomes more industrialized, stress-related diseases, suicides, divorce and other social problems are likely to rise.
- Environmental costs: A richer society may be more concerned for the environment, but it is also likely to do more damage to it. The higher the level of consumption, the higher is likely to be the level of pollution and waste. Acid rain and the depletion of the ozone layer have been two examples.
- Non-renewable resources: If growth involves using a greater amount of resources, rather than using the same amount of resources more efficiently, certain non-renewable resources will run out more rapidly.

Macroeconomic objectives: Macroeconomics is concerned with issues, objectives and policies that affect the whole economy. All economic analysis that refers to aggregate is macro. The unemployment rates, the inflation rate, the rate of economic growth are all aggregate and therefore macro issues.

The four macro major objectives are:

1. Full employment
2. Price stability
3. A high, but sustainable, rate of economic growth
4. Keeping the balance of payment in equilibrium

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FULL EMPLOYMENT OR LOW UNEMPLOYMENT: The claimant count is the older, more out-of-date, measure of unemployment used in the UK. Those counted must be out of work, physically able to work and looking for it, and actually claiming benefit.

For a more realistic count, and for international comparisons, the ILO (international labor organization) measure is used. This includes the young unemployed who are not always eligible to claim, married women who can't claim if their husband is earning enough, and those who claim sickness and invalidity benefits.

Only those who are active are included in the working population (or labor force), which is defined as all those who are employed to registered unemployed. But some of the inactive are in this category by choice, for instance, students and those who retire early.

It is important to keep unemployment levels as low as possible. High unemployment is expensive for the government and, therefore, for the taxpayer. For every unemployed person, there are two costs to the government. First, the unemployed worker will be entitled to benefit, and if he/she is young, or older but remains unemployed for a long period of time, he/she should be offered training. Secondly, there is the less obvious cost of the loss of income tax revenues the workers would have paid in work. These workers would have been paying VAT as well through their purchases.

There are other costs of unemployment. There is the cost of whole economy in terms of wasted, unused resources. The existence of any idle resources means that the economy will be at a point within its production possibility frontier (PPF). Unemployed workers (young men, in particular) may create other external costs in the economy, like crime for example. Finally, there is the personal human cost to each worker. In the short term the unemployed worker has to put up with the loss of earnings, although this may be balanced by redundancy payments. But in the long run, the long term unemployed will find it harder and harder to find a job, as they find that the skills they have become less relevant and they have had no new training.

PRICE STABILITY: Inflation is usually defined as a sustained rise in the general level of prices. Technically it is measured as the annual rate of change of the retail price index RPI, often referred to as the headline rate of inflation. For prices to be stable, therefore, the inflation rate should be zero. Generally, governments are happy if they can keep the inflation rate down to a low percentage.

Some would say that the main reason why the control of inflation is so important is that if inflation gets out of control, the economy stops growing. If inflation rises, the central bank is forced to raise interest rates. Consumers will stop borrowing to spend and firms will stop borrowing to invest. The housing market will slump, and along with it all the home improvement consumption that goes with it. Manufacturer's exports will become less competitive. The economy may well drop into a recession.

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Many economists worry about the re-distributional effects of inflation. Periods of high inflation cause redistribution from savers to borrowers. Inflation erodes the real value of all money.

The interest rate that is earned on one's saving will usually be higher than the inflation rate, but there have been times when the rate of interest does not keep up with the inflation rate, and so savers are paid a negative real interest rate.

Most of your parents have made money on their house. Of course, they do not realize (i.e. get their hands on) this gain in wealth whilst they live in the property, but their mortgage remains relative constant (especially if they are on a fixed rate mortgage) as their earnings rise over the years. Also, they can borrow against the increased value of the property to buy a car, go on holiday or extend the house. Economists do not like this redistribution from savers to borrower. It penalizes thrift, which is a bad thing for the economy because, over the long term, the amount of investment in an economy is closely related to the amount of saving.

There are two minor costs of inflation. Shoe leather cost refers to the time wasted (and worn shoe leather) searching the market place for the lowest price. This is much harder when inflation is high. High inflation tends to coincide with variable inflation and, therefore, very unstable prices.

Menu costs refer to the cost of inflation to businesses in terms of continually having to change their menus, price tags, vending machines, etc. due to the continually changing price.

HIGH (but sustainable) ECONOMIC GROWTH: Economic growth tends to be measured in terms of the rate of change of real GDP (gross domestic product). When the word real accompanies any statistic, it means that the effects of inflation have been removed. GDP is a measure of the annual output (or income, or expenditures) of an economy.

Some would say this is the most important of all the objectives. Why? Since economic growth leads to improved living standards. Obviously this is good for the inhabitants of an economy.

Unfortunately, there always seems to be a trade-off between efficiency and equity in an economy. The most efficient economies in the world have grown the most of their inhabitants have experienced the best standards of living. But these economies tend to have the most unequal distributions of income. The competitive world of the free market creates winners and losers. The winners are exceptionally rich, but the losers can be worse off in absolute terms as well as relative terms unless the government's welfare state is generous.

To be fair, though, successful economies tend to grow so much that the standard of living of the poorest household in the economy still tends to rise. This was known as the trickle-down effect.

As the national cake gets bigger, the poor may well be getting a smaller slice (worsening distribution of income), but that slice will still constitute more cake as time goes by.

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BALANCE OF PAYMENTS IN EQUILIBRIUM: It records all flows of money into, and out of a country over a given time period (usually a year). It is split in to two: the current account and the capital and financial accounts.

Probably the most important is the current account because this records how well the country is doing in terms of its exports of goods and services relative to its imports.

Japan has the largest current account surplus in the world. Although a surplus sounds better than a deficit, both can be bad. Japan's surplus forces other countries in the world to have deficits. In fact, while Japan's surplus is the biggest in the world, the USA's deficit is the biggest in the world. This is not a coincidence! The UK tends to be in deficit, although the current account was in surplus a couple of years ago, mainly due to its strength in the service sector.

For developed economies with mature and free capital markets, balance of payments disequilibrium are not so important nowadays. If a developed country has, for example, a current account deficit, it can usually attract enough foreign investment on the capital account to balance their books. In other words, they can finance the deficit. Some economists argue that it is a bad thing, but more in the sense that it is a sign of the uncompetitiveness of the economy's industries. Economies with continual current account deficits are not 'paying their way' in the world.

The problem lies with developing countries that experience large current account deficits over a long period of time. They may simply not be able to finance this deficit. They need to pay for the imports in the currency of the country from whom the goods are bought. The US dollar is usually acceptable worldwide. If they can't earn enough US dollars to pay for their imports, can't attract the foreign investment and can't convince other countries to lend them money, then they will be in trouble. Countries in this much trouble can knock on the door of the International Monetary Fund (IMF) looking for the funds. These will only be lending under quite severe economic conditions, like keeping government spending down to a prohibitively low level.

Current account surpluses might not seem so bad (I'd prefer my bank account to be in surplus rather than deficit!), but they do mean that the economy is sacrificing consumption at home for exports abroad. Also, they can be bad for the simple reason that one country's surplus is another country's deficit. It is probably best not to ruin a country to which you export. Also one doesn't want to sour trade relations; otherwise you might provoke a trade war.

In the 1960s, the balance of payments was considered very important. A deficit was considered highly embarrassing in those days. The long-term sustainability of a deficit was a big problem in the days before global free movements of capital. Deficits would reduce the demand for the domestic currency relative to other currencies, and so the value of the currency against other currencies would fall. This was unacceptable within the 'Bretton Woods fixed exchange rate system'. Nowadays, with a floating/managed ER systems and high global capital flows, many economists

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believe that this balance of payments deficits or surpluses (on currency account) simply do not matter.

Growth and low inflation have always been important. Without growth, people's standards of living will not increase, and if inflation is too high then the value of money falls negating any increase in living standards. Nowadays these are definitely the two most important objectives of macroeconomic policy of many countries.

Are there any conflicts between these objectives? Unfortunately, it is virtually impossible for a government to score in all these goals at once. We shall begin with the three major conflicts and then look at two more that are linked to microeconomics.

1. **Healthy growth and low inflation:** If an economy grows too quickly, especially if it is due to excessive consumer spending then demand will outstrip supply and prices will rise. Equally, the steps taken to keep inflation low, like relatively high interest rates can often restrict growth via reduced consumer spending and investment. It is difficult to achieve both aims. The trend rate of both is seen as the rate of growth an economy can grow without igniting inflation. Most economists believe that this is around 2% to 3% at the moment.
2. **Low unemployment (or full employment) and low inflation:** This is the classic conflict in economic theory. In fact, an economist called Phillips constructed a curve using empirical data to show that this conflict existed (although this does not mean that the relationship would hold forever).
These two variables have, in theory, an inverse relationship. If a government tries to reduce unemployment through reflationary measures, such as lower interest rates or increased public spending, then the resulting reduction in unemployment will push wages, and then prices higher. On the other hand, when the government tries to control high inflation with higher interest rates and reduced spending, the resulting reduced consumer spending and lower investment will result in job losses.
3. **Healthy growth and the environment:** Of course, not every time would consider the environment a 'minor' objective, but unfortunately governments have not quite woken up to the problem yet. Although there have been summits at which controls on various types of pollution were agreed, the US amongst others seem to find it difficult to keep their promises! Quite simply, the faster the rate of growth, the higher the level of production, and so the level of pollution from factories, cars etc. rises. Also, vital rain forest tends to disappear, not just because we consume the wood; new factories, towns and housing are built in resulting land.
4. **Healthy growth and equality:** Equality was an objective of socialist governments and so is now obsolete in the world of new labor. Although it is true to say that forcing equality throughout a country can lead to inefficiencies. The developed world has grown hugely since the Second World War, but even with the creation of welfare states it is the wealth creators that have benefited hugely whilst those at the bottom of the pile have seen their

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standard of living just plod along. The communist Soviet Union kept the more equality conscious socialist model going right into the 80s, but its inefficiencies meant that the rate of growth was much slower. Now even they have embraced capitalism, although the transition has not exactly been smooth!

Inflation and unemployment: In 1958 A.W Phillips explained a tradeoff between inflation and unemployment. He argued that government adopts expansionary policies by increasing AD, then AS takes time to adjust which puts pressure in the price level of rise. So an increase in AD leads to a reduction in unemployment at the cost of increased rates of inflation. He explained this relation with the help of Phillips curve which is downward sloping and intersects the horizontal axis at the natural rate of unemployment, after intersecting the horizontal axis the Phillips curve flattens out which means that as the unemployment rise above the natural level, the price level or the wage rate fall very slowly or don't fall at all i.e. wages are said to be sticky downwards. On the other hand if the government tries to reduce the unemployment below the natural level it will lead to inflation in the economy.

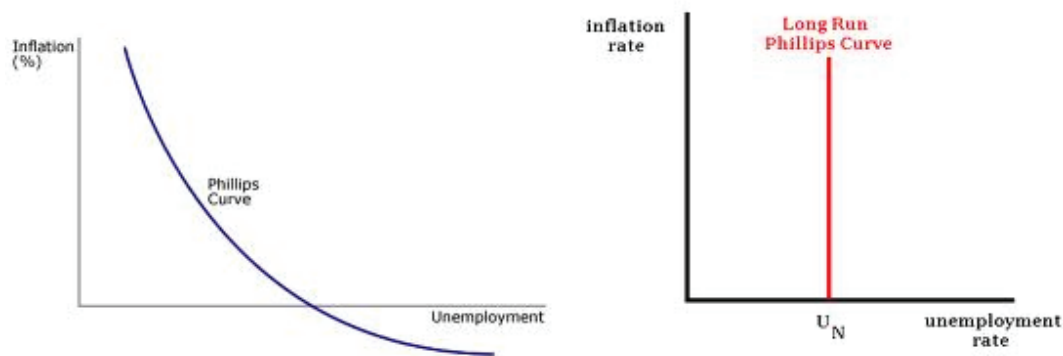
In the 1960-70s the Phillips curve relationship was jeopardized when inflation and unemployment increased at the same time, a term called slump-inflation. Keynes defended the Phillips curve relationship by explaining the cause of simultaneous increase in inflation and unemployment which was external shock of increase in world oil prices which increased the cost of production all over the world causing the price level to rise along with the increase in unemployment.

Monetarist attacked the Phillips curve relationship and argued that an increase in AD as a result of expansionary policy will reduce the natural rate of unemployment for a short period. They said that workers will raise their expectations about the rate of inflation which will cause their real wage or purchasing power to fall. So they will bargain for higher wages and will not be deceived by money illusion. This increase in wage rate will lead to an increase in cost of production and producers will reduce the output level causing unemployment to rise back to the natural level and rate of inflation stays at the same level. This creates a vertical long run Phillips curve or expectation augmented Phillips curve, so according to monetarists there is no relationship between inflation and unemployment in the long-run.

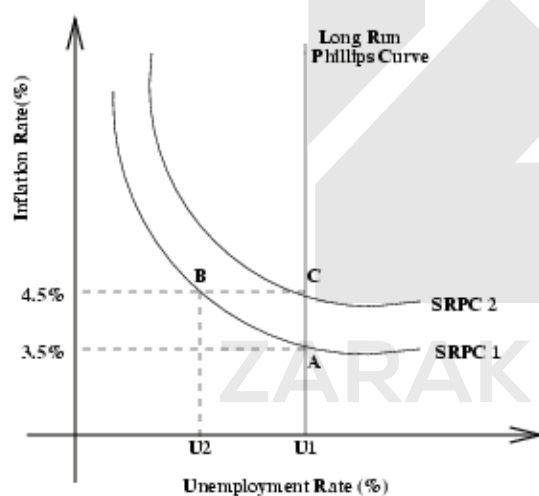
Money illusion is an economic theory positing that people have a tendency to view their wealth and income in nominal dollar terms, rather than in real terms. Money illusion is sometimes also referred to as price illusion.

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Philips theories provided a distinction between short-run and long-run scenarios. To explain the inadequacies of the Phillips Curve he and Friedman developed the Expectation-Augmented Phillips Curve which combines the inflationary expectation and the concept of natural rate of unemployment. They claimed that in the long-run the expected inflation adjust to changes in actual inflation and in the short-run the Philips curve shifts to the right. This means that in the long run there is no inverse relationship between the inflation rate and the unemployment rate, and the Phillips Curve is a vertical line at the natural rate of unemployment. In the short-run, however, the Phillips Curve shifts to the northeast, as we can see in Figure 2. In the figure, U_1 and U_2 represent the natural rate of unemployment at two short-run Phillips curves (SRPC) and the vertical line represents the long-run Phillips Curve (LRPC).



Keynesians and monetarist

Keynesians: 1950s and 1960s Keynesians argued for demand management policies, i.e. for the government to control AD. If the economy grew too fast, the government deflated; if it grew too slowly it reflatd through fiscal policy. This led to a series of stop-go policies, e.g. deflate, reflate, and deflate. Keynesians believe that markets do not clear and are slow to adjust, e.g. the labor

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market. This means the economy can settle in equilibrium below full employment, so cyclical or involuntary unemployment exists. The government should intervene to stabilize the economy. Fiscal policy is more effective than monetary policy. Inflation is often caused by cost push factors. Extreme Keynesians believe markets do not clear and the economy will not move towards full employment. The government must expand demand.

Monetarist: In the 1970s the UK experienced stagflation (high inflation and high unemployment). Existing theories struggled to explain it; monetarism grew in appeal. Monetarism is based on the quantity theory of money, $MV = PT$. V and T are assumed constant so prices are directly related to money supply.

Monetarist believes the inflation is due to the money supply growing faster than output growth. Reducing the rate of growth of the money supply will lead to less inflation without more unemployment (in the long run) prices and wages change quite quickly, so the economy tends towards full employment. Inflation makes firms uncompetitive, discourages investment, and so governments must control inflation. To do this they must control the money supply. Apart from this, the government should intervene very little.

Extreme monetarists are called the new classical school - they believe markets clear quickly and expectations adapt very quickly. Faster growth of the money supply will quickly lead to inflation; even in the short run the Phillips curve is vertical.

Moderate monetarists believe markets adjust fairly quickly. An increase in money supply and therefore demand will lead to some fall in unemployment in the short run. Similarly, a sharp reduction in the money supply may cause unemployment in the short run.

Unemployment: The meaning of unemployment: Unemployment can be expressed either as a number or as a percentage. The most usual definition that economists use for the number unemployed is those between the ages 18 and 60 who are without work, but who are available for work at current wage rate. If the figure is to be expressed as a percentage then it is the percentage of the total labor force. The labor force is defined as those in employment plus those unemployed. Thus, if 22.5m people are employed and 2.5m people were unemployed, the unemployment rate would be

Unemployment rate = (Registered unemployed/labor force) x 100

$$= \{2.5/(22.5+2.5)\} \times 100 = 10\%$$

Official measures of unemployment

Claimant count: Two common measures of unemployment are used in official statistics. The first is the claimant count unemployment which includes anyone between the ages of 18 and 60 receiving

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unemployment benefits such as job seekers allowance. Claimant statistics have the advantage of being very easy to collect. However, they exclude all those of working age who are available for work at current wage rate, but who are not eligible for benefits e.g. married women with husbands earning sufficient income, people in their 50s and 60s receiving pensions, people seeking part-time work rather than full time work. The claimant count thus understates the true level of unemployment.

Recognizing the weakness of claimant count, UK government since 1998 has used the standardized unemployment rate, as the main measure of unemployment. In this measure the unemployed are defined as people of working age who are without work, available to start work and actively seeking employment or waiting to take up an appointment.

Types of unemployment

1. Frictional unemployment: this occurs when people leave their jobs either voluntarily or because they are sacked or made redundant and are unemployed for a period of time while they are looking for a new job. They may not get the first job they apply for, despite a vacancy existing. The employer may continue searching, hoping to find a better qualified person. Likewise unemployed people may choose not to take the first job they are offered. Instead they may continue searching, hoping that a better one will turn up. The problem is that information is imperfect. Employers are not fully informed about what labor is available; workers are not informed about what jobs are available. Employers therefore, have to search for the right labor and workers searching for the right jobs. One obvious remedy for frictional unemployment is to provide better job information. This could be provided by government job centers, by private employment agencies or by local and national newspapers. Another much more controversial remedy is for the government to reduce the level of unemployment benefits. This will make the unemployed more desperate to get a job and thus prepared to accept a lower wage.
2. Structural unemployment: It is where the structure of the industry changes. Employment in some industries may expand while in others it contracts. There are two main reasons for this:
 - A change in the pattern of demand: some industries experience declining demand. This may be due to change in consumer tastes. Certain goods go out of fashion. Or it may be due to competition from other industries. For example, consumer demand may shift away from coal to other fuels. This will lead to structural unemployment in mining areas.
 - A change in the method of production (technological unemployment): new techniques of production often allow the same level of output to be produced with fewer workers. This is known as labor saving technical progress. Unless output expands sufficiently to absorb the surplus labor, people will be made redundant. This creates technological unemployment.

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Structural unemployment often occurs in particular regions of the country. When it does, it is referred to as regional unemployment. It is due to the concentration of particular industries in particular areas. There are two broad approaches to tackle structural unemployment i.e. market oriented and interventionist. A market oriented approach involves encouraging people to look more actively for jobs, if necessary in other parts of the country. It involves encouraging people to adopt a more willing attitude towards retraining and if necessary to accept some reduction in wages. An interventionist approach involves direct government action to provide jobs to the unemployed. Two examples are providing grants to firms to set up in areas of high unemployment (regional policy) and government funded training schemes.

3. Seasonal unemployment: It occurs when demand for certain type of labor fluctuates with the seasons of the year. This problem is particular in holiday/tourism industries or areas where unemployment can reach very high in winter months.
4. Demand deficient or cyclical unemployment: This type of unemployment is associated with economy in recession. As the economy moves into recession, consumer demand falls. Firms find that they are unable to sell their current level of output. For a time they may be prepared to build up stock of unsold goods, but sooner or later they will start to cut back on production and cut back on the amount of labor they employ. The deeper the recession becomes and the longer it lasts, the higher the demand deficient unemployment becomes. As the economy recovers and begins to grow again the demand deficient unemployment starts to fall again. Because demand deficient unemployment fluctuates with the business cycles, it is sometimes referred to as cyclical unemployment.

The non accelerating inflation rate of unemployment (NAIRU)/Natural rate of unemployment

It is the rate of unemployment which is consistent with a given rate of inflation i.e. it does not affect the level of inflation. It is the sum of frictional, structural and seasonal unemployment which makes about 1 to 3% of the total labor force. This type of unemployment always exists in every economy and nothing much can be done to reduce this type of unemployment.

NAIRU is the rate of unemployment which exists when the economy is producing its potential output i.e. where the economy is operating at full employment and there is no cyclical unemployment. At this level of unemployment the demand for labor is equal to the supply of labor at the existing real wage rate. Even when there is no general disequilibrium in the economy at the current wage levels - even when there are as many job vacancies as people unemployed - there will still be some unemployment. For various reasons, not all vacancies will be filled. The problem is one of mismatching. There may be excess demand for labor (vacancies) in some markets and excess supply (unemployment) in others. There may be vacancies for computer technicians and unemployment in the steel industry, but unemployed steel workers cannot immediately become computer technicians.

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There may even be vacancies and unemployment in the same market, but the unemployed may be unable, unwilling or unsuitable to take up the vacancies or simply unaware of their existence.

Costs of unemployment: The most obvious cost of unemployment is to the unemployed themselves. There is the direct financial cost of the loss in their earnings measured as the difference between their previous wage and unemployment benefit. Then there are personal costs of being unemployed. The longer people are unemployed, the more dispirited they may become. Their self esteem is likely to fall, and they are more likely to succumb to stress related diseases.

Then there are costs to the family and friends of the unemployed. Personal relations become strained and there may be an increase in domestic violence and the number of families splitting up. Then there are broader costs to the economy; unemployment represents a loss of output. In other words actual output will be less than potential output. Apart from the lack of income to the unemployed themselves, this underutilization of resources leads to lower income for other people too:

- The government loses tax revenues, since the unemployed pay no income tax and given that the unemployed spend less, they pay less indirect tax (sales tax). The government also incurs administrative costs associated with the running of benefits offices. It may also have to spend extra on health care, the social services and the police.
- Firms lose the profits that could have been made, had there been full employment
- Other workers lose any additional wages they could have earned from higher national output.
- The longer people remain unemployed, the more de-skilled they tend to become, thereby reducing potential as well as actual income.
- Finally, there is some evidence that higher unemployment leads to increased crime and vandalism. This obviously imposes a cost on the sufferers.

Benefits: The costs of unemployment may sometimes be offset by the benefits of unemployment. If the workers voluntarily quit their jobs to look for a better one, then they must realize that benefits of a better job more than compensate their temporary loss of income. From the nation's point of view, a workforce that is prepared to quit jobs and spend a short time unemployed will be a more adaptable, more mobile workforce that is responsive to changing economic circumstances. Such a workforce will lead to a greater allocative efficiency in the short run and more rapid economic growth in the long run.

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